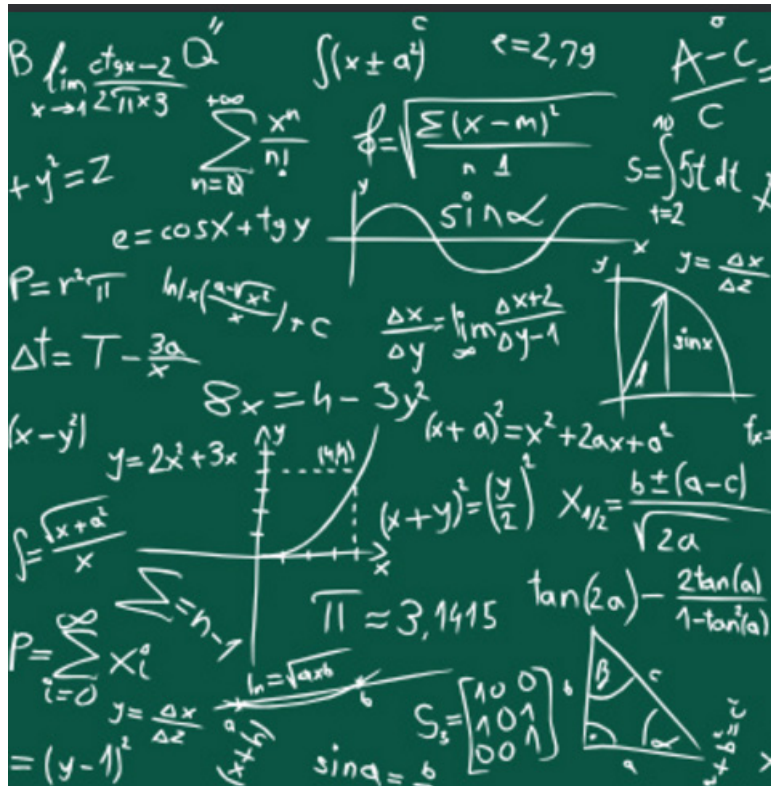


Exercises: Maclaurin expansions



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

Exercise 1

Using the Maclaurin expansion of the following functions $f(x)$, determine the equation of the tangent line to the graph at the point $(0, f(0))$ and the local position of the graph with respect to this tangent

$$f(x) = 1 + \sin(\sqrt{1+x^3} - 1)$$

$$f(x) = 2\sqrt{1 + \sinh x} - x$$

$$f(x) = 8\sqrt{1 + \sin x} + x^2$$

$$f(x) = 8\sqrt{1 + \log(1+x)} + 3x^2$$

$$f(x) = 8\sqrt{1 - 2x - x^2} - 8 + 8x$$

From

$$\sqrt{1+x^3} - 1 = \frac{1}{2}x^3 + o(x^3)$$

e

$$\sin y = y - \frac{1}{6}y^3 + o(y^3)$$

we have

$$f(x) = 1 + \sin(\sqrt{1+x^3} - 1) = 1 + \frac{1}{2}x^3 + o(x^3).$$

The equation of the tangent line is

$$y = 1.$$

The term $\frac{1}{2}x^3 + o(x^3)$ says that there exists a neighborhood $(-\delta, \delta)$, $\delta > 0$, of $x = 0$ where the graph crosses the tangent line from below for $-\delta < x < 0$ to above for $0 < x < \delta$. The point $x = 0$ is an inflection point with horizontal tangent.

From

$$\sinh x = x + o(x^2)$$

e

$$2\sqrt{1+y} = 2 + y - \frac{1}{4}y^2 + o(y^2)$$

we have

$$f(x) = 2\sqrt{1+\sinh x} - x = 2 + x - \frac{1}{4}x^2 + o(x^2) - x = 2 - \frac{1}{4}x^2 + o(x^2).$$

The equation of the tangent line is

$$y = 2.$$

The term $-\frac{1}{4}x^2 + o(x^2)$ says that there exists a neighborhood $(-\delta, \delta)$, $\delta > 0$, of $x = 0$ where the graph lies below the tangent line for $-\delta < x < \delta$, $x \neq 0$. In particular, since the tangent line is horizontal, the point $x = 0$ is a local maximum point.

From

$$\sin x = x - \frac{1}{6}x^3 + o(x^3)$$

e

$$8\sqrt{1+y} = 8 + 4y - y^2 + \frac{1}{2}y^3 + o(y^3)$$

we have

$$f(x) = 8\sqrt{1+\sin x} + x^2 =$$

$$8 + 4x - \frac{2}{3}x^3 - \left(x - \frac{1}{6}x^3\right)^2 + \frac{1}{2}\left(x - \frac{1}{6}x^3\right)^3 + o(x^3) + x^2 =$$

$$8 + 4x - \frac{2}{3}x^3 - x^2 + \frac{1}{2}x^3 + x^2 + o(x^3) = 8 + 4x - \frac{1}{6}x^3 + o(x^3).$$

The equation of the tangent line is

$$y = 8 + 4x.$$

The term $-\frac{1}{6}x^3 + o(x^3)$ says that there exists a neighborhood $(-\delta, \delta)$, $\delta > 0$, of $x = 0$ where the graph crosses the tangent line from above for $-\delta < x < 0$ to below for $0 < x < \delta$. The point $x = 0$ is an inflection point.

From

$$\log(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + o(x^3)$$

e

$$8\sqrt{1+y} = 8 + 4y - y^2 + \frac{1}{2}y^3 + o(y^3)$$

we have

$$f(x) = 8\sqrt{1+\log(1+x)} + 3x^2 =$$

$$8 + 4\left(x - \frac{1}{2}x^2 + \frac{1}{3}x^3\right) - \left(x - \frac{1}{2}x^2 + \frac{1}{3}x^3\right)^2 +$$

$$\frac{1}{2}\left(x - \frac{1}{2}x^2 + \frac{1}{3}x^3\right)^3 + o(x^3) + 3x^2 =$$

$$8 + 4x - 2x^2 + \frac{4}{3}x^3 - x^2 + x^3 + \frac{1}{2}x^3 + 3x^2 + o(x^3) =$$

$$8 + 4x + \frac{17}{6}x^3 + o(x^3).$$

The equation of the tangent line is

$$y = 8 + 4x.$$

The term $\frac{17}{6}x^3 + o(x^3)$ says that there exists a neighborhood $(-\delta, \delta)$, $\delta > 0$, of $x = 0$ where the graph crosses the tangent line from below for $-\delta < x < 0$ to above for $0 < x < \delta$. The point $x = 0$ is an inflection point.

Exercise 2

Write the Maclaurin expansion of order 3 of the function

$$f(x) = e^x - e^{-x^2} - \sin x$$

and determine its order of infinitesimal. Based only on the expansion obtained, say whether the function has at $x = 0$ a relative minimum point, a relative maximum point, an inflection point, or none of these. Justify your answer.

Since we have to expand $f(x)$ to the third order, we can stop at the second order in the expansion of e^{-x^2} . That is:

$$e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + o(x^3)$$

$$e^{-x^2} = 1 - x^2 + o(x^3)$$

$$\sin x = x - \frac{1}{6}x^3 + o(x^3)$$

We then have

$$\begin{aligned} f(x) &= 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + o(x^3) - (1 - x^2 + o(x^3)) - \left(x - \frac{1}{6}x^3 + o(x^3)\right) \\ &= \frac{3}{2}x^2 + \frac{1}{3}x^3 + o(x^3) \end{aligned}$$

In particular, we have

$$f(x) \sim \frac{3}{2}x^2 \quad \text{as } x \rightarrow 0$$

from which we can immediately state that the order of infinitesimal of $f(x)$ is $\alpha = 2$ and the principal part is $\frac{3}{2}x^2$. The point $x = 0$ is a local minimum point; indeed, the term $\frac{3}{2}x^2 + o(x^2)$ says that there exists a neighborhood $(-\delta, \delta)$, $\delta > 0$, of $x = 0$ where the graph lies entirely above the horizontal tangent line $y = 0$.

Exercise 3

Determine the Taylor polynomial of order 3, centered at the point $x_0 = \frac{\pi}{3}$, of the function $f(x) = \cos x$.

In general, the Taylor polynomial of order n , centered at x_0 , of a function $f(x)$ is

$$T_{n,f,x_0}(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

with $f^{(0)}(x_0) = f(x_0)$. In this case, we need the first, second and third derivatives, evaluated at $x_0 = \frac{\pi}{3}$. We have

$$f'\left(\frac{\pi}{3}\right) = -\sin\left(\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

$$f''\left(\frac{\pi}{3}\right) = -\cos\left(\frac{\pi}{3}\right) = -\frac{1}{2}$$

$$f'''\left(\frac{\pi}{3}\right) = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

and the Taylor polynomial of order 3, centered at the point $x_0 = \frac{\pi}{3}$, of the function $f(x) = \cos x$ is

$$T_{3,f,\frac{\pi}{3}}(x) = \frac{1}{2} - \frac{\sqrt{3}}{2} \left(x - \frac{\pi}{3}\right) - \frac{1}{4} \left(x - \frac{\pi}{3}\right)^2 + \frac{\sqrt{3}}{12} \left(x - \frac{\pi}{3}\right)^3$$